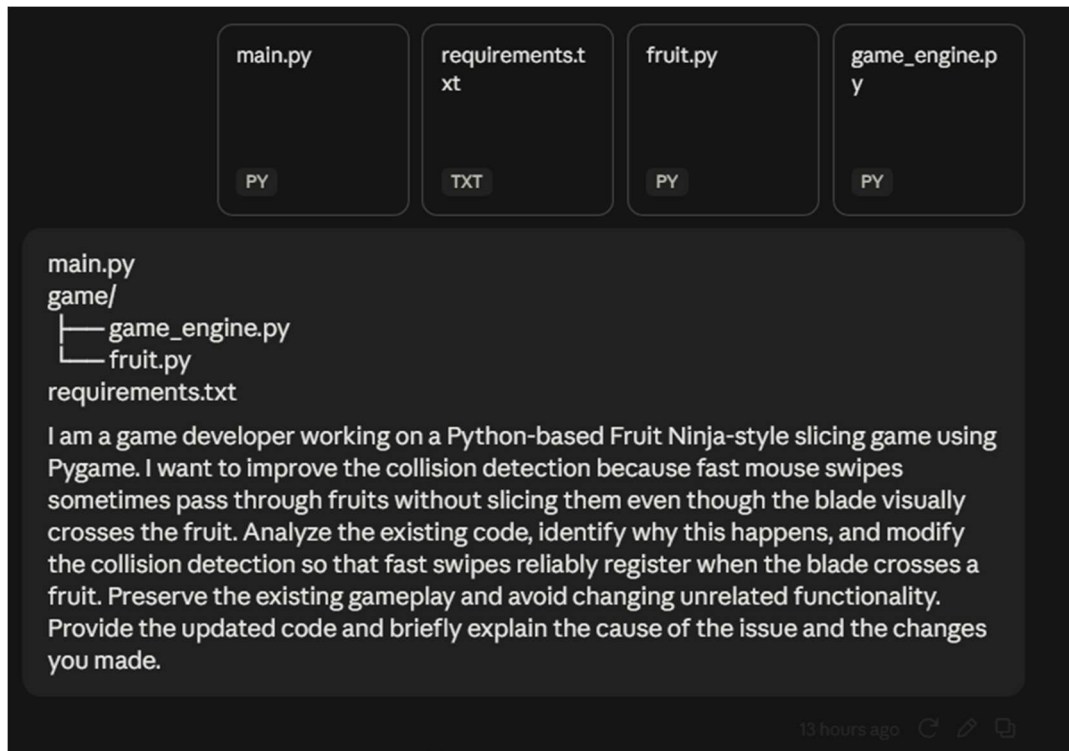


Software Engineering

Lab 4: VibeCoding

Prompts

Task 1



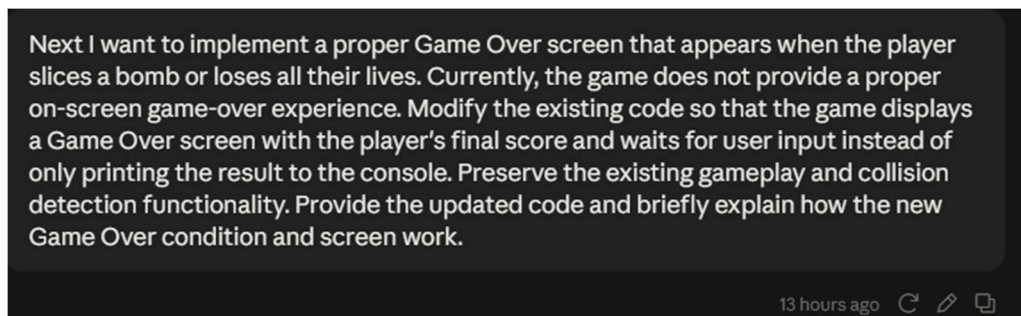
The screenshot shows a VibeCoding interface with a dark theme. At the top, there is a file explorer with four files: `main.py` (PY), `requirements.txt` (TXT), `fruit.py` (PY), and `game_engine.py` (PY). Below the file explorer is a code editor. The editor shows a file tree on the left with `main.py` selected, and the main area contains the following text:

```
main.py
game/
├── game_engine.py
├── fruit.py
└── requirements.txt
```

I am a game developer working on a Python-based Fruit Ninja-style slicing game using Pygame. I want to improve the collision detection because fast mouse swipes sometimes pass through fruits without slicing them even though the blade visually crosses the fruit. Analyze the existing code, identify why this happens, and modify the collision detection so that fast swipes reliably register when the blade crosses a fruit. Preserve the existing gameplay and avoid changing unrelated functionality. Provide the updated code and briefly explain the cause of the issue and the changes you made.

At the bottom right of the code editor, there is a timestamp "13 hours ago" and three icons: a refresh icon, a pencil icon, and a copy icon.

Task 2



The screenshot shows a VibeCoding interface with a dark theme. The code editor contains the following text:

Next I want to implement a proper Game Over screen that appears when the player slices a bomb or loses all their lives. Currently, the game does not provide a proper on-screen game-over experience. Modify the existing code so that the game displays a Game Over screen with the player's final score and waits for user input instead of only printing the result to the console. Preserve the existing gameplay and collision detection functionality. Provide the updated code and briefly explain how the new Game Over condition and screen work.

At the bottom right of the code editor, there is a timestamp "13 hours ago" and three icons: a refresh icon, a pencil icon, and a copy icon.

Task 3

Now, I want to add a replay option after the Game Over screen. The player should be able to choose between Easy, Medium, and Hard difficulty levels, with the difficulty affecting the fruit spawn rate or bomb chance, or choose to exit the game. When a new game starts, the score and lives should reset properly. Integrate this feature with the existing Game Over system without breaking the current gameplay. Provide the updated code and briefly explain how the difficulty selection, replay, and exit options work.

13 hours ago   

Task 4

ok and the last task is that I want to add basic sound feedback to improve the gameplay experience. The game should play a sound when a fruit is sliced, a different sound when a bomb is hit, and another sound when the game ends. Integrate the sounds into the existing game without affecting the current gameplay or previously implemented features. Provide the updated code, mention any additional sound files or dependencies required, and briefly explain where and when each sound effect is triggered.

13 hours ago   